Flood situation report

Area: Chamoli rock-ice avalanche and flood, Uttarakhand, 7 Feb 2021 | Event date: 7 February 2021 | Latest radar image: 15 February 2021

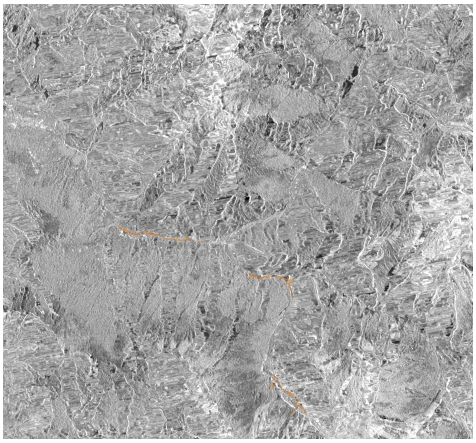
1. Where it flooded

Satellite images show 137 ha (1.37 km^2) flooded or buried by debris. F010 F011

Of this, 8.9 ha is open water and 128 ha is debris or scour. F012 F014

The radar could not see 7,961 ha because of terrain shadow, so conditions there are unknown. F016

The places with the most buildings likely destroyed: Bhangyul. F042



Radar change map, before and after the event.

2. What was hit

Buildings: 1 likely destroyed and 1 possibly damaged, out of 237 mapped. F025 F021 F022

Roads: 1.1 km of road lies under mapped debris or water; the road sections that touch it total 1.3 km, out of 240.8 km mapped. F039 F033 F037

Bridges: 1 likely destroyed and 0 possibly damaged, out of 13 mapped. F031 F027 F028

3. Who is cut off

2 of 58 settlements are likely cut off by road. F044 F043

Among them: Bhangyul, Garghaniya. F048

The nearest health facility for 2 of the cut-off settlements is Joshīmath. F049 F050

OpenStreetMap lists 4 health facilities in the area (hospitals, health posts and clinics); its labels are not verified. F404 F405

Repair priority

1. Reopening unnamed service bridge reconnects 2 settlements. F407 F408 F411

Method

Sentinel-1 radar images from before and after the event are compared to map new water and debris. Pre-event OpenStreetMap roads, bridges and buildings are overlaid, and a road network model finds settlements that can no longer reach a health facility.

Educational prototype, not an operational tool. Numbers come from satellite analysis and have not been checked on the ground.

Grey codes such as F012 point to entries in facts.json.

Contains modified Copernicus Sentinel data 2026.
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